

## DURANGO, Mexico

WMO: 764230

Lat: 24.0614N

Lon: 104.6006W

Elev: 1872

StdP: 80.77

Time Zone: -6.00 (MEX)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-----|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |     |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |     |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p) |
| 1             | 0.5        | 2.4 | -10.0                         | 2.0 | 13.6 | -8.2 | 2.3 | 12.4 | 10.5                  | 14.7 | 7.9 | 19.2 | 0.1                   | 50   | N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 6             | 14.3                   | 33.7            | 16.8 | 32.3 | 16.4 | 31.0 | 15.9 | 21.1                | 28.0 | 19.8 | 26.9 | 19.0 | 26.5 | 2.4                  | 140  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.9                            | 17.3 | 25.4 | 17.6 | 15.9 | 23.7 | 16.7 | 15.0 | 22.1 | 70.4          | 28.1 | 65.7 | 27.1 | 62.4 | 26.4 | 27.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |  |  |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|--|--|
|                   |      |     | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |  |  |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |  |  |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |  |  |
| 7.2               | 6.0  | 5.0 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
|                   |      |     | DB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
|                   |      |     | WB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 18.7   | 12.5  | 15.4  | 17.2  | 20.3  | 22.6  | 23.7  | 21.8  | 21.8  | 20.5  | 18.9  | 15.6  | 13.6 |
|                                                                                  | DBStd       | 4.24   | 2.58  | 2.48  | 2.62  | 2.47  | 2.21  | 2.10  | 1.51  | 1.66  | 1.66  | 2.28  | 2.44  | 2.92 |
|                                                                                  | HDD10.0     | 17     | 7     | 1     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 8    |
|                                                                                  | HDD18.3     | 586    | 181   | 84    | 50    | 12    | 3     | 0     | 0     | 1     | 3     | 21    | 84    | 148  |
|                                                                                  | CDD10.0     | 3185   | 86    | 152   | 226   | 309   | 390   | 412   | 367   | 365   | 315   | 276   | 169   | 118  |
|                                                                                  | CDD18.3     | 711    | 0     | 3     | 16    | 71    | 134   | 162   | 108   | 107   | 68    | 38    | 3     | 0    |
|                                                                                  | CDH23.3     | 7112   | 13    | 129   | 372   | 972   | 1609  | 1579  | 693   | 761   | 453   | 387   | 107   | 35   |
|                                                                                  | CDH26.7     | 2366   | 0     | 11    | 67    | 334   | 719   | 720   | 172   | 199   | 84    | 53    | 6     | 1    |
| Wind                                                                             | WSAvg       | 1.7    | 1.6   | 2.0   | 2.2   | 2.3   | 2.1   | 1.7   | 1.6   | 1.6   | 1.3   | 1.2   | 1.3   | 1.5  |
| Precipitation                                                                    | PrecAvg     | 321    | 10    | 5     | 1     | 6     | 17    | 43    | 76    | 78    | 69    | 17    | 9     | 14   |
|                                                                                  | PrecMax     | 607    | 55    | 38    | 6     | 39    | 133   | 146   | 201   | 169   | 209   | 86    | 87    | 44   |
|                                                                                  | PrecMin     | 100    | 0     | 0     | 0     | 0     | 0     | 1     | 1     | 0     | 10    | 0     | 0     | 0    |
|                                                                                  | PrecStd     | 161    | 13    | 10    | 1     | 9     | 29    | 40    | 64    | 49    | 57    | 20    | 18    | 13   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 25.0  | 28.1  | 30.3  | 32.7  | 35.2  | 35.6  | 31.9  | 32.0  | 30.6  | 29.7  | 27.5  | 25.8 |
|                                                                                  |             | MCWB   | 11.7  | 13.3  | 14.3  | 14.2  | 17.3  | 17.4  | 19.3  | 18.2  | 18.1  | 16.8  | 15.7  | 15.0 |
|                                                                                  | 2%          | DB     | 23.2  | 26.5  | 28.6  | 31.3  | 33.6  | 34.0  | 30.2  | 30.2  | 28.7  | 28.2  | 25.8  | 24.2 |
|                                                                                  |             | MCWB   | 11.3  | 13.3  | 13.1  | 13.7  | 16.0  | 17.5  | 18.4  | 18.1  | 17.7  | 16.2  | 14.4  | 13.3 |
|                                                                                  | 5%          | DB     | 21.8  | 25.0  | 27.1  | 30.2  | 32.1  | 32.7  | 28.9  | 28.9  | 27.5  | 27.1  | 24.5  | 22.9 |
|                                                                                  |             | MCWB   | 10.6  | 12.7  | 12.7  | 13.4  | 15.7  | 17.0  | 18.1  | 17.9  | 17.6  | 16.2  | 13.5  | 12.1 |
|                                                                                  | 10%         | DB     | 20.3  | 23.4  | 25.6  | 28.9  | 30.7  | 31.2  | 27.5  | 27.6  | 26.3  | 25.8  | 23.1  | 21.4 |
|                                                                                  |             | MCWB   | 10.0  | 12.2  | 12.4  | 13.0  | 14.9  | 17.0  | 17.9  | 17.9  | 17.5  | 15.5  | 13.0  | 11.2 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 14.5  | 21.8  | 22.2  | 16.0  | 20.2  | 22.0  | 21.8  | 21.5  | 21.4  | 20.7  | 18.9  | 17.0 |
|                                                                                  |             | MCDB   | 22.1  | 24.7  | 25.3  | 29.8  | 31.7  | 32.7  | 30.1  | 28.6  | 26.7  | 26.8  | 24.7  | 24.5 |
|                                                                                  | 2%          | WB     | 12.5  | 18.4  | 18.2  | 14.9  | 18.0  | 19.6  | 20.1  | 19.9  | 19.5  | 18.3  | 15.9  | 14.3 |
|                                                                                  |             | MCDB   | 20.8  | 21.5  | 21.5  | 29.4  | 29.8  | 29.2  | 27.7  | 27.0  | 25.8  | 25.0  | 23.3  | 22.7 |
|                                                                                  | 5%          | WB     | 11.5  | 14.8  | 14.4  | 14.1  | 16.7  | 18.8  | 19.1  | 19.1  | 18.7  | 17.3  | 14.7  | 12.8 |
|                                                                                  |             | MCDB   | 20.0  | 21.4  | 23.7  | 28.5  | 29.4  | 28.1  | 26.6  | 26.3  | 25.1  | 24.3  | 22.1  | 21.2 |
|                                                                                  | 10%         | WB     | 10.6  | 12.6  | 13.0  | 13.3  | 15.8  | 18.0  | 18.3  | 18.4  | 18.1  | 16.6  | 13.6  | 11.7 |
|                                                                                  |             | MCDB   | 19.0  | 21.5  | 23.9  | 27.3  | 28.1  | 27.2  | 25.5  | 25.6  | 24.4  | 23.4  | 21.0  | 20.1 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 16.0  | 16.2  | 16.5  | 17.1  | 16.7  | 14.3  | 11.8  | 12.2  | 11.4  | 14.5  | 15.6  | 16.0 |
|                                                                                  |             | MCDBR  | 17.9  | 18.1  | 18.6  | 18.6  | 18.6  | 16.6  | 13.9  | 14.2  | 14.1  | 15.8  | 17.5  | 17.8 |
|                                                                                  |             | MCWBR  | 9.1   | 9.0   | 8.1   | 7.2   | 6.9   | 5.4   | 4.8   | 5.1   | 5.3   | 6.4   | 8.0   | 9.1  |
|                                                                                  | 5% WB       | MCDBR  | 16.2  | 17.2  | 16.7  | 17.7  | 17.1  | 13.8  | 12.7  | 12.3  | 11.7  | 13.4  | 15.8  | 16.8 |
|                                                                                  |             | MCWBR  | 9.0   | 12.1  | 9.8   | 7.2   | 7.3   | 5.2   | 5.3   | 5.2   | 5.0   | 6.1   | 7.9   | 9.5  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.260  | 0.276 | 0.270 | 0.292 | 0.346 | 0.355 | 0.374 | 0.361 | 0.360 | 0.310 | 0.273 | 0.263 |      |
|                                                                                  | taud        | 2.632  | 2.535 | 2.574 | 2.512 | 2.386 | 2.412 | 2.417 | 2.460 | 2.445 | 2.583 | 2.642 | 2.639 |      |
|                                                                                  | Ebn at Noon | 1021   | 1018  | 1034  | 1007  | 943   | 927   | 910   | 926   | 925   | 968   | 997   | 1005  |      |
|                                                                                  | Edh at Noon | 83     | 98    | 99    | 108   | 123   | 119   | 118   | 113   | 112   | 93    | 82    | 80    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 4.58   | 5.60  | 6.75  | 7.48  | 7.46  | 6.63  | 6.01  | 6.02  | 5.38  | 5.49  | 4.90  | 4.36  |      |
|                                                                                  | RadStd      | 0.34   | 0.47  | 0.28  | 0.45  | 0.42  | 0.48  | 0.33  | 0.33  | 0.39  | 0.33  | 0.34  | 0.25  |      |

## Historical Trends

|                            | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |         |         |         |
|----------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|---------|---------|---------|
|                            |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                            |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only          | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (1 neighbor) | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | +0.64 | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page